

Phase 1 Technical Report

System Modelling and Open-Loop Validation of a Quadcopter UAV

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1. Introduction

Unmanned Aerial Vehicles (UAVs), particularly quadcopters, have become increasingly important in applications such as surveillance, inspection, mapping, search and rescue, and autonomous navigation. The performance of these systems depends heavily on their ability to maintain stability and accurately track desired attitudes under varying operating conditions.

Before designing any control strategy, it is essential to develop an accurate mathematical representation of the system. A controller can only perform effectively if the model used during simulation closely reflects the actual physical behavior of the vehicle. For this reason, the first phase of the project focuses entirely on modelling and validating the rotational dynamics of the quadcopter.

Objective

Develop a nonlinear rotational dynamics model using the Newton-Euler formulation and implement it in MATLAB. The model is then validated through open-loop simulations to ensure that the generated responses are physically correct and suitable for future controller development.

Foundation for Future Phases

The validated model developed in this phase will serve as the foundation for all subsequent stages of the project, including PID controller design, Sliding Mode Control implementation, comparative performance analysis, and eventual hardware deployment.

2. System Description

A quadcopter is a multi-rotor aerial platform consisting of four propellers arranged symmetrically around a central frame. By varying the speeds of individual motors, the vehicle generates thrust and control torques that enable motion about three rotational axes.

The three primary rotational motions are:



Roll (φ)

Roll represents rotation about the longitudinal axis of the vehicle. This motion causes one side of the quadcopter to rise while the opposite side moves downward.



Pitch (θ)

Pitch represents rotation about the lateral axis. It causes the front of the vehicle to move upward or downward relative to the rear.



Yaw (ψ)

Yaw represents rotation about the vertical axis and controls the heading direction of the quadcopter.



For attitude control applications, understanding the relationship between applied torques and rotational motion is essential. This relationship is captured through the dynamic model developed in this phase.

3. Mathematical Modelling

The rotational dynamics of the quadcopter are derived using the Newton-Euler rigid body formulation. This approach is widely used in aerospace systems because it accurately captures rotational motion while remaining computationally efficient for simulation and control design.

The rotational motion of a rigid body can be expressed as:

$$I\dot{\omega} = \tau - \omega \times (I\omega)$$

Variables

- I – Inertia matrix
- ω – Angular velocity vector
- τ – Applied torque vector

The second term accounts for gyroscopic and coupling effects that arise when the vehicle rotates simultaneously about multiple axes.

Simplifying Assumptions

To simplify the analysis, the quadcopter is assumed to possess a symmetric frame structure. Under this assumption:

- Roll and pitch inertias are equal
- The inertia matrix remains diagonal
- Cross-axis inertia terms are neglected

Resulting Rotational Equations

Roll Dynamics

$$I_{xx}\ddot{\varphi} = \tau_{\varphi} + (I_{yy} - I_{zz})\dot{\theta}\dot{\psi}$$

Pitch Dynamics

$$I_{yy}\ddot{\theta} = \tau_{\theta} + (I_{zz} - I_{xx})\dot{\varphi}\dot{\psi}$$

Yaw Dynamics

$$I_{zz}\ddot{\psi} = \tau_{\psi} + (I_{xx} - I_{yy})\dot{\varphi}\dot{\theta}$$

These equations form the core mathematical model used throughout the project. The nonlinear coupling terms are particularly important because they capture interactions between different rotational axes. Although these effects may appear small during near-hover conditions, they become increasingly significant during aggressive maneuvers and external disturbances.

4. State-Space Representation

MATLAB numerical solvers require the system to be represented in first-order state-space form. Therefore, the second-order rotational equations are converted into a set of six first-order differential equations.

The state vector is defined as:

$$x = [\varphi, \dot{\varphi}, \theta, \dot{\theta}, \psi, \dot{\psi}]^T$$

φ — Roll Angle	$\dot{\varphi}$ — Roll Rate
θ — Pitch Angle	$\dot{\theta}$ — Pitch Rate
ψ — Yaw Angle	$\dot{\psi}$ — Yaw Rate

- ☐ The resulting state-space model consists of **six states** and **three control inputs** corresponding to roll, pitch, and yaw torques. This representation enables direct integration using MATLAB's ODE45 solver and provides a suitable framework for future controller implementation.

5. MATLAB Implementation

The developed mathematical model was implemented in MATLAB using a modular programming approach. This structure improves readability, simplifies debugging, and allows future phases to reuse the same model without modification.

The implementation consists of three primary files.

1	2	3
Parameter Definition Module The parameter file stores all physical properties required by the simulation, including vehicle mass, arm length, roll inertia, pitch inertia, yaw inertia, thrust coefficient, drag coefficient, and gravitational acceleration. In addition to storing system parameters, the script calculates several useful quantities such as total vehicle weight, hover thrust, and estimated hover motor speed. Sanity checks are included to verify that all parameter values remain within physically realistic limits.	Dynamics Module The dynamics function implements the nonlinear Newton-Euler equations derived earlier. The function accepts the current state vector, applied control torques, and system parameters, and returns the state derivatives required by the numerical solver. This module represents the mathematical plant that will later be controlled using PID and Sliding Mode Control algorithms.	Simulation Module The simulation script performs numerical integration of the state equations using MATLAB's ODE45 solver. Tight solver tolerances are selected to accurately capture the small nonlinear coupling effects present within the model. The simulation module is also responsible for generating plots, comparing analytical and numerical results, and performing model validation.

6. Open-Loop Validation

Once the mathematical model was implemented, a series of open-loop tests were conducted to verify its correctness. The purpose of these tests was to ensure that the model accurately reproduces expected physical behavior before introducing any controller.

Test Case 1: Pure Roll Motion

A constant roll torque was applied while all initial states were set to zero. Under these conditions, the quadcopter experiences constant angular acceleration about the roll axis.

The simulation response was compared against the analytical solution obtained from rigid body rotational dynamics. The resulting roll angle followed the expected parabolic trajectory, while the roll rate increased linearly with time.

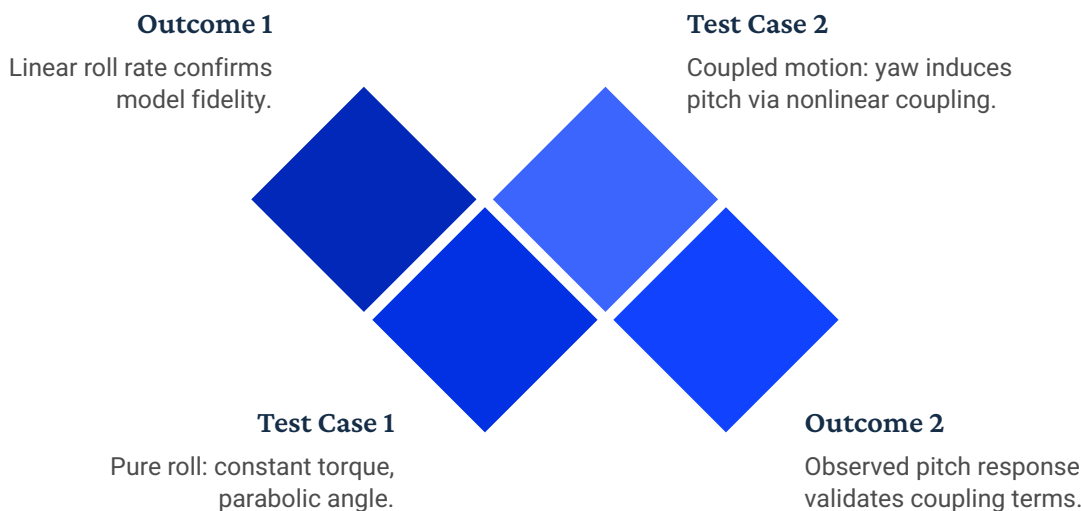
This close agreement confirmed that the rotational equations and numerical implementation were correct.

Test Case 2: Coupled Rotational Motion

The second validation test introduced an initial yaw rate while maintaining the same roll torque input.

Although no pitch torque was applied, a pitch response gradually emerged during the simulation. This behavior occurred because of the nonlinear coupling term present in the pitch dynamics equation.

The observed response demonstrated that the model successfully captures gyroscopic interactions between rotational axes and accurately represents nonlinear system behavior.



The two open-loop test cases progressively validate the model, first confirming basic rotational behavior and then verifying nonlinear cross-axis coupling effects.

7. Results and Discussion

The simulation results confirmed the validity of the developed model.



Test Case 1 — Excellent Analytical Agreement

The first test case demonstrated excellent agreement between analytical predictions and numerical simulation results. The roll response behaved exactly as expected for a rigid body subjected to constant torque.



Test Case 2 — Nonlinear Coupling Validated

The second test case successfully revealed the influence of nonlinear coupling effects. The appearance of a pitch response despite zero pitch torque validated the presence and implementation of gyroscopic interactions within the model.

These observations indicate that the developed model accurately captures both primary rotational behavior and cross-axis interactions.



Control Perspective: From a control perspective, the results highlight why advanced control methods become necessary for UAV applications. As operating conditions become more dynamic, nonlinear effects can no longer be neglected, and controllers must be capable of handling system uncertainties and coupling phenomena.

8. Conclusion and Future Work

A nonlinear rotational dynamics model of a quadcopter UAV was successfully developed using the Newton-Euler formulation and implemented in MATLAB.

Phase 1 Complete

The model was validated through open-loop simulations and analytical comparisons. The results demonstrated accurate representation of rotational motion and successfully captured nonlinear coupling effects between axes.

Simulation Platform

The successful completion of this phase establishes a reliable simulation platform for future controller development.

Next Phase

In the next phase, a PID-based attitude controller will be integrated with the developed model to achieve closed-loop stabilization and reference tracking. The controller performance will then serve as a baseline for comparison with the Sliding Mode Controller developed in later stages of the project.

The validated model obtained in this phase will remain unchanged throughout the remainder of the project and will act as the plant for all subsequent simulations and performance evaluations.